Dylan Thornburg ECEN 115 LAB2

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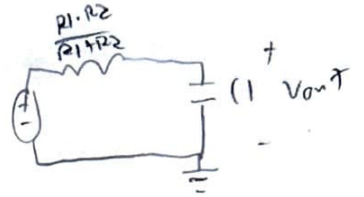


$V_{in}\left(\frac{R2}{R1+R2}\right)$

$$V_{out}(t) = V_{in}\left(\frac{R2}{R1+R2}\right) + \left[V_0 - V_{in}\left(\frac{R2}{R1+R2}\right)\right] e^{\frac{-t}{T}}$$

where $T = \left(\frac{R1 \cdot R2}{R1+R2}\right) \cdot C$

a.

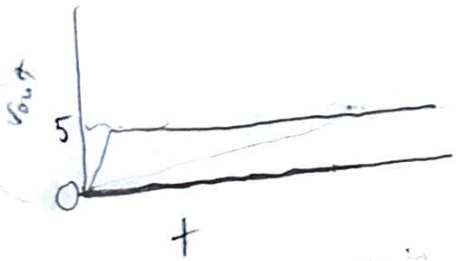
$$V_{out}(t) = V_{in}\left(\frac{R2}{R1+R2}\right) + \left[V_0 - V_{in}\left(\frac{R2}{R1+R2}\right)\right] e^{\frac{-t}{\left(\frac{R1R2}{R1+R2}\right)\cdot C}}$$

b.

when
$C1 = .1\mu F$ and
$V_0 = 0$

$$V_{out} = \frac{1}{2} V_{in} - \frac{1}{2} V_{in} e^{-1000t}$$

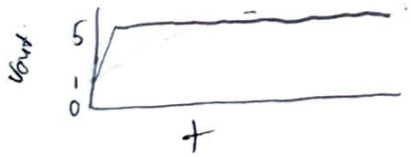
when $V_{in} = 0$, $V_{out} = 0$
when $V_{in} = 10$, $V_{out} = 5 - 5e^{-1000t}$



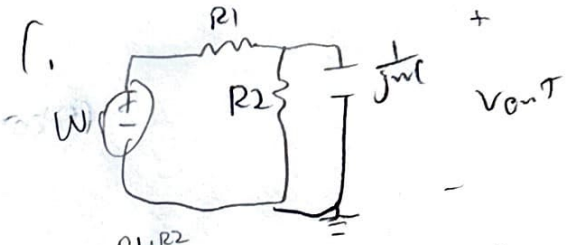
when
$C1 = .1\mu F$ and
$V_0 = 1$

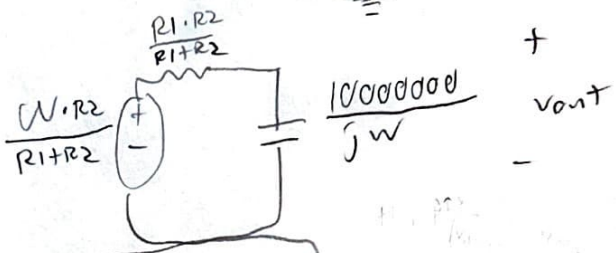
$$V_{out} = \frac{1}{2} V_{in} + \left[1 - \frac{1}{2} V_{in}\right] e^{-1000t}$$

when $V_{in} = 0$, $V_{out} = 1$
$V_{in} = 10$, $V_{out} = 5 - 4e^{-1000t}$

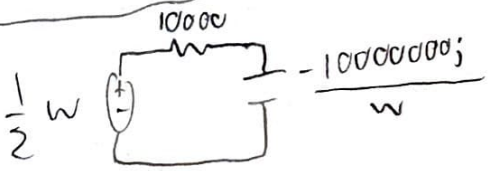


c.





i. low pass filter



$$\text{cutoff freq} = \frac{1}{RC} = \frac{1}{10000 \cdot 10^{-6} \cdot .1} = 1000$$

ii. $W = 1000$ rad/s is the cutoff freq when converted to Hz, $f = 159.15$ Hz

$$H(jw) = \frac{V_{out}}{V_{in}} = \frac{\frac{1}{jwC}}{10000 + \frac{1}{jwC}} = \frac{1}{10000jwC + 1}$$

$$\frac{V_{out}}{V_{in}} = \frac{1}{.001jw + 1} \qquad \left|\frac{V_{out}}{V_{in}}\right| = \frac{1}{\sqrt{(.001w)^2 + 1}}$$

at $w = 0$, gain $= 1$
at $w = \infty$, gain $= 0$